GURDHARAM JEET SINGH

Software Engineer Intern • Full-Stack & Web Systems

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EDUCATION

Baba Farid Group of Institutions (BFGI) — **Expected May 2028**

B.Sc. (Hons.) Computer Science • ***CGPA: 8.4 / 10.0*** — *Punjab, India*

Coursework: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Computer Networks, Operating Systems

Senior Secondary Examination (Class XII, PSEB) — 91.0% — **Completed May 2024**

Non-Medical Stream (Physics, Chemistry, Mathematics) — *Punjab, India*

Matriculation Examination (Class X, PSEB) — 89.2% — **Completed May 2022**

General Sciences, Mathematics & Computer Applications — *Punjab, India*

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL (PostgreSQL), HTML5, CSS3, Git, Bash

Frontend: React, Next.js, Tailwind CSS, Vite, Three.js, WebGL, Canvas API, Framer Motion, Responsive Design

Backend & Systems: Node.js, Express, PostgreSQL, REST APIs, WebSockets, Docker, OOP, Testing (Pytest, Jest)

AI & Cloud Tools: LLMs, AI Agents, Gemini API, Meta Cloud API (WABA), GitHub Actions CI/CD, Railway

PROJECTS

EduCore — College Management ERP & Student Information System — **Jan 2025 – Present**

React, TypeScript, Express, PostgreSQL, MariaDB Connection Pools, Railway — github.com/gurination1/ERP_educore

- **Campus Administration & Academic Management:** Built full-stack college ERP managing student admissions, academic grading, and fee reconciliation across campus departments.
- **Database Abstraction & RBAC:** Implemented pooled PostgreSQL and MariaDB database abstraction with automated connection health checks and 9-role JWT access controls.

yt-auto — 24/7 Autonomous Cloud Media Production Fleet — **Feb 2025 – Present**

Python, FFmpeg, GitHub Actions CI/CD Matrix, Gemini Multimodal AI — github.com/gurination1/yt-auto

- **Distributed Cloud Processing:** Built automated media pipeline orchestrating batch video generation across parallel GitHub Actions matrix runners with zero dedicated server costs.
- **Autonomous Quality Control Loop:** Automated video rendering with custom FFmpeg filtergraphs, integrating a Gemini AI evaluation check that automatically verifies visual frames and triggers re-renders.

DoodHisaab — Offline-First Dairy Procurement & Ledger Mobile App — **Apr 2024 – Oct 2024**

Flutter, Dart, SQLite (sqflite), Riverpod State Management, Workmanager — github.com/gurination1/doodhisaab

- **Offline-First Transaction Ledger:** Built production Flutter mobile app enabling rural dairy farmers to record transactions offline with zero data loss, utilizing local SQLite storage and Riverpod state management.
- **Resilient Mobile Architecture:** Engineered background synchronization with Workmanager, dynamic milk-fat rate calculations, and shipped signed release APKs on GitHub.

Creative & Spatial Web Suite — Branders, Dream Heights & SOLUM Minerals — **May 2025 – Present**

Next.js, Three.js, WebGL/GLSL, JavaScript, Tailwind CSS, Canvas API, Lenis Scroll — github.com/gurination1/branders

- **Procedural 3D & WebGL Shaders:** Built custom Three.js WebGL viewport (tablet-3d.js) with procedural canvas textures (SOLUM Minerals) and GLSL fluid shaders (Branders), rendering dynamic lighting at locked 60 FPS.
- **Modern Performance Engineering:** Developed agency flagships with Next.js and Lenis momentum scroll, eliminating layout shifts (CLS 0.00) and reducing asset payload by 85% via procedural synthesis.

WORK EXPERIENCE

CP Systems (Cool & Pure System) — **Jan 2024 – Jan 2025**

Software Developer (Contract / Part-Time) • Industrial RO Plants & Dairy Equipment (cpsystem.in) — *Punjab, India*

- **Industrial Telemetry & Database Optimization:** Built operational telemetry and inventory microservices using Node.js and SQL, optimizing query plans to reduce p95 latency by 42%.
- **Meta Cloud API & CI/CD Pipelines:** Automated customer notifications over Meta Cloud API and implemented GitHub Actions CI/CD with caching, reducing build times from 14m to 3.8m.

Gurdharam Enterprise — **Jan 2025 – Present**

Freelance Full-Stack Developer • Web Platforms & Client Engineering — *Remote*

- **Commercial Client Delivery:** Developed production web applications, agency portals (Branders), and responsive frontend interfaces, delivering sub-second page loads.
- **Conversational Voice Pipeline:** Built low-latency voice pipeline integrating Exotel WebSockets, Silero VAD voice interruption, and streaming Sarvam AI Indic LLMs.